

Joel Teply

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Dear Hiring Team,

I am applying for DFA's AI Enterprise Architect position. I am a Kansas native, a University of Kansas Computer Engineering graduate, and a hands-on software and AI architect with more than twenty years of experience building production systems across machine learning, distributed systems, cloud infrastructure, native platforms, and enterprise software.

This role is particularly interesting to me because it sits at the intersection where I have spent much of my career: taking ambitious AI capabilities and designing the architecture required to make them scalable, reliable, secure, and usable in production.

I co-founded Cambrian and served first as CEO and later CTO and Principal Engineer, remaining deeply hands-on in architecture and implementation. We built computer vision and machine learning technology for enterprise customers across retail, home improvement, and automotive. I worked with customers from discovery and feasibility through system architecture, integration, debugging, optimization, deployment, and delivery. Our technology was licensed by major companies, reached millions of users, and operated across applications, SDKs, web platforms, cloud services, and enterprise integrations.

The AI systems behind those products required architecture well beyond the models themselves. I built production Python and TensorFlow services, APIs, S3-backed data and ML pipelines, Docker-based CPU/GPU deployments, automated testing, scalable AWS infrastructure, model tooling, and high-performance native inference systems. I have designed systems that combine deep learning with real-time state, tracking, geometry, multi-frame fusion, and constrained on-device execution, as well as the cloud and data infrastructure surrounding them.

My current work on Continuum extends that experience into modern AI and distributed architectures. I am the sole architect of a large Rust platform for persistent multimodal AI agents spanning local inference, tool execution, persistent

memory and state, scheduling, evaluation, continual learning, distributed compute, peer-to-peer communication, and heterogeneous hardware. I have also worked directly with model deployment and optimization through LoRA, quantization, structured pruning, mixed precision, model routing, and evaluation.

A major focus of that work is what happens when AI moves beyond a demonstration and becomes infrastructure. Multiple models and agents compete for finite compute and memory; inference and training have to move across heterogeneous machines; learned artifacts need provenance; failures have to be diagnosable; and evaluation needs to determine whether adaptations are actually improvements. Those are architecture problems as much as AI problems.

That perspective is what I would bring to DFA. I am comfortable translating an ambiguous business requirement into a durable technical design, defining boundaries and integration patterns, evaluating technologies and tradeoffs, and then working closely enough with implementation teams to make sure the architecture survives contact with production. My background spans APIs, event-driven and distributed systems, databases, cloud infrastructure, security-sensitive SDKs, ML pipelines, GPU execution, edge systems, and the application layers that ultimately consume those capabilities.

I also understand that enterprise AI architecture requires governance as part of the design rather than as an afterthought. My recent AI work emphasizes reproducible evaluation, provenance, controlled model adaptation, persistent state, and measurable performance. Earlier work in biometric authentication and cryptographic systems gave me direct experience building software where security, privacy, identity, and trust are fundamental architectural concerns.

While the specific enterprise platforms vary between organizations, I have spent my career learning complex systems quickly and making architectural decisions across unfamiliar boundaries. I would be comfortable working with DFA's Azure, data, ERP, CRM, security, and integration teams to establish AI patterns that fit the broader enterprise rather than creating a separate AI island.

I also value that this opportunity is here in Kansas. I grew up here, earned my Computer Engineering degree at KU, and built my company and career in the Kansas City area. The opportunity to apply that experience to DFA's enterprise AI architecture while remaining deeply engaged with the technical work is especially compelling to me.

Thank you for your consideration. I would welcome the opportunity to discuss DFA's AI architecture goals and the technical challenges involved in building a scalable enterprise AI capability.

Sincerely,

Joel Teply